Basic Inferential Statistics with R

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Chapter box

Central question: How can we reproduce a basic estimate, confidence interval or statistical test in R and explain its estimand, assumptions and bounded interpretation?
Focal concepts: Sampling and working models; reference distributions; SEs, intervals and tests for means, proportions and associations; paired and independent comparisons; one-way ANOVA and focused comparisons.
Main evidence: Paired measurements for ten patients in `sleep` continue Chapter 5. `mtcars` and `ToothGrowth` provide separate basic-method examples, and a short declared Normal simulation makes sampling variability visible.
Scope: Chapters 3-4 supply inferential foundations and Chapter 5 supplies descriptive calculations and simple R. Chapter 1 supplies the reading approach; Chapter 1a remains optional. Statistics organizes the route. Chapter 7 develops regression; optional appendices retain technical depth.

Core route. Sections 1–3 carry the paired `sleep` example from its recorded differences to one inferential answer. Sections 4–5 then show how the same question–estimand–procedure discipline changes for other basic methods. Section 6 uses repeated sampling to check what an inferential procedure does, and Section 7 completes the bounded answer and bridges to regression. This route is complete without the appendices; they provide optional mathematical, computational, and design extensions.

1 From a Descriptive Result to an Inferential Question

Chapter 5 described ten people's drug-2-minus-drug-1 differences in extra sleep: a mean of 1.58 hours and an SD of 1.23 hours. The descriptive arithmetic concerns those recorded people. An interval concerns an additional estimand under a stated sampling or working model. A descriptive population question can therefore require inference; these are analytical roles, not competing question-mode labels.

1.1 Recall the paired observations

The sleep documentation defines extra sleep relative to control under two drugs. `ID` identifies the person, `group` the condition and `extra` the hours. The conditions are not baseline and follow-up. Recreate the ten aligned pairs locally:

```
sleep_wide <- tidyr::pivot_wider(sleep, id_cols = ID,
  names_from = group, values_from = extra, names_prefix = "drug_")
complete_pair <- complete.cases(sleep_wide[c("drug_1", "drug_2")])
sleep_wide <- sleep_wide[complete_pair, ]
```

```r
difference <- sleep_wide$drug_2 - sleep_wide$drug_1
c(n = length(difference), mean = mean(difference), SD = sd(difference))
```

```
    n  mean    SD
10.00  1.58  1.23
```

All ten pairs are complete. Their mean is exactly calculable from the records. The working inferential question is: **What do these differences estimate about a process mean, if they behave as independent observations from the stated process?** The file does not itself establish a sampling, treatment-order or carryover protocol. Keep that provenance separate from the model supporting a reference distribution.

1.2 A declaration comes before a default

A formula such as `score ~ group` names an outcome and grouping variable. It does not state whether observations are independent or paired. Begin with the question and dependence structure, then choose the calculation.

Table 1
Questions guide the calculation.

Question	Relevant quantity and calculation	Condition to establish
Mean paired difference	Mean of within-person differences; paired or one-sample analysis	Correct pairs; independence between people for ordinary inference
Mean difference between separate groups	Difference of means; Welch comparison	Independent observations within and across groups
Fraction meeting a criterion	Proportion and suitable interval	Stated denominator and appropriate repeated-trial model
Association between categories	Counts, association measure and reference test	Independent units; adequate reference conditions
Differences across several groups	Group means, omnibus comparison or contrast	Stated design, variance and multiplicity conditions

This is an orientation, not an automatic selection algorithm. Sections 4 and 5 work through the additional procedures. Functions return structured objects; their printed summaries are only one view. We will extract the quantities relevant to a question after understanding the calculation.

2 Probability and Reference Distributions

A probability calculation describes a stipulated model. It supplies tail areas and critical values for an inferential procedure when that model is appropriate. Use direct R calls and read their statistical question before the function name. Sections 4-5 apply these references to additional records.

2.1 Four prefixes, four probability questions

Prefix	Returned quantity	Example
`d`	Discrete probability mass or continuous density height	`dbinom()`, `dnorm()`, `dt()`
`p`	Cumulative probability	`pbinom()`, `pnorm()`, `pt()`
`q`	Quantile at a cumulative probability	`qbinom()`, `qnorm()`, `qt()`
`r`	Random draws from the specified model	`rbinom()`, `rnorm()`, `rt()`

For $X \sim \text{Binomial}(20, .30)$, the model describes 20 independent trials with a common success probability .30. These are not 20 observations discovered in either additional dataset.

```
mass_at_4 <- dbinom(4, size = 20, prob = .30)
cumulative_at_4 <- pbinom(4, size = 20, prob = .30)
percentile_90 <- qbinom(.90, size = 20, prob = .30)
c(mass = mass_at_4, cumulative = cumulative_at_4,
  percentile = percentile_90)
```

```
      mass cumulative percentile
    0.1304     0.2375     9.0000
```

```
dnorm(0)
```

```
[1] 0.3989
```

```
pnorm(1) - pnorm(-1)
```

```
[1] 0.6827
```

```r
qnorm(.975)
```

```
[1] 1.96
```

`dbinom(4, 20, .30)` is $P(X = 4)$. For a continuous normal variable, `dnorm(0)` is a density height, not $P(X = 0)$; the latter is zero. An interval probability is an area, here $\Phi(1) - \Phi(-1) \approx .6827$. A continuous density can exceed one without assigning a probability above one. The binomial quantile is the smallest integer whose cumulative probability reaches the requested level; discrete quantiles need not attain that probability exactly.

2.2 Normal, t, chi-square, and F references

For an independent normal sample, replacing known σ by its sample estimate in $(\bar{Y} - \mu)/(\sigma/\sqrt{n})$ changes the exact reference from standard normal to t_{n-1}. The heavier t tails reflect this estimated scale. This is not a universal cutoff saying that every sample above some size is normal or independent.

```r
c(normal = qnorm(.975), t_5 = qt(.975, 5),
  t_9 = qt(.975, 9), t_30 = qt(.975, 30))
```

```
normal    t_5    t_9   t_30
 1.960  2.571  2.262  2.042
```

```r
qchisq(.95, df = 4)
```

```
[1] 9.488
```

```r
qf(.95, df1 = 2, df2 = 40)
```

```
[1] 3.232
```

```r
pchisq(10, df = 4, lower.tail = FALSE)
```

```
[1] 0.04043
```

```r
pf(4, df1 = 2, df2 = 40, lower.tail = FALSE)
```

```
[1] 0.02608
```

A chi-square variable is a sum of squared independent standard normals. If $U \sim \chi^2_a$ and $V \sim \chi^2_b$ independently, then $(U/a)/(V/b) \sim F_{a,b}$. This explains the two degrees of freedom of an F reference. Pearson count discrepancies can have an approximate chi-square reference; normal linear-model mean-square comparisons can have an exact F reference. The same distribution name does not establish the same conditions for every statistic using it.

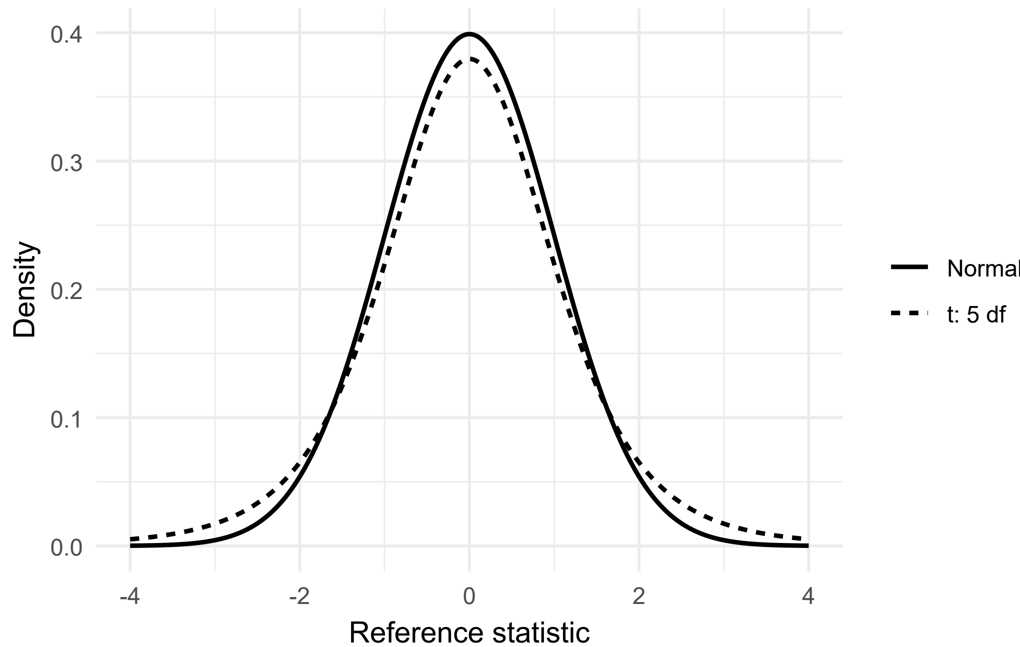

Figure 1: Standard normal and t densities. The heavier t tails produce larger finite-df critical values.

3 Estimate a Mean Difference and Its Uncertainty

3.1 Introduce a different question when uncertainty is needed

The ten-record mean is already known. For a **working-model question**, suppose the differences behave as independent draws from a stable process with mean μ_D and variance σ^2_D. What do these observations estimate about that process mean?

This changes the estimand from the average of named records to a process parameter. The short historical data description does not itself establish random sampling, treatment order or absence of carryover. Keep those design questions separate from the following conditional calculation.

For independent normally distributed differences, the one-sample t interval is exact under the model. For other independent distributions it is an approximation whose adequacy depends on shape and sample size. Here $n = 10$, so normality is a consequential assumption. Pairing accounts for dependence within a person; independence concerns differences between people. It is the differences, not both marginal condition distributions, to which the normal model applies.

3.2 Execute the arithmetic before reading output

The SD of differences is

$$s_D = \sqrt{\frac{\sum_i (d_i - \bar{d})^2}{n-1}} = 1.2300.$$

Under the independent-differences model,

$$\widehat{SE}(\bar{D}) = \frac{s_D}{\sqrt{n}} = \frac{1.2300}{\sqrt{10}} = 0.3890 \text{ hours.}$$

The 95% interval uses $t_{.975,9} = 2.2622$:

$$\bar{d} \pm t_{.975,9}\frac{s_D}{\sqrt{n}} = 1.5800 \pm 2.2622(0.38896) = [0.7001,\ 2.4599] \text{ hours.}$$

```
n <- length(difference)
estimate <- mean(difference)
s <- sd(difference)
se <- s / sqrt(n)
critical <- qt(.975, df = n - 1)
interval <- estimate + c(-1, 1) * critical * se
interval
```

```
[1] 0.7001 2.4599
```

The endpoint calculation uses unrounded values. For $H_0 : \mu_D = 0$,

$$t = \frac{\bar{d} - 0}{s_D/\sqrt{n}} = \frac{1.58}{0.38896} = 4.0621, \qquad df = 9.$$

The two-sided p value is about .00283 under the stated null model. It is the probability of a statistic at least as extreme under that model, not the probability that the null is true. The interval concerns a mean difference, not a range containing 95% of individual differences.

```r
result <- t.test(sleep_wide$drug_2, sleep_wide$drug_1,
                 paired = TRUE, mu = 0, conf.level = .95)
result$estimate
```

```
mean difference
           1.58
```

```r
result$stderr
```

```
[1] 0.389
```

```r
result$conf.int
```

```
[1] 0.7001 2.4599
attr(,"conf.level")
[1] 0.95
```

```r
result$statistic
```

```
    t
4.062
```

```r
result$parameter
```

```
df
 9
```

```r
result$p.value
```

```
[1] 0.002833
```

R subtracts the second vector from the first: drug 2 minus drug 1 here. `paired = TRUE` uses within-person differences, but does not match IDs for you. The preceding pivot provided that alignment.

The one-sample call below gives the same estimate, interval and test:

```r
difference_result <- t.test(difference, mu = 0, conf.level = .95)
difference_result$estimate
```

```
mean of x 
     1.58 
```

```r
difference_result$conf.int
```

```
[1] 0.7001 2.4599
attr(,"conf.level")
[1] 0.95
```

This is a useful computational identity: a paired t procedure is a one-sample procedure on differences. Agreement verifies the calculation, while its model and design still need their own justification.

3.3 Direction, confidence level and test sidedness

Reversing the subtraction reverses the estimate and both interval endpoints; it does not change the two-sided p-value. A higher confidence level uses a larger critical value and gives a wider interval on the same data. The alternative hypothesis must be specified to match the question before examining the observed sign:

```r
t.test(difference, mu = 0, alternative = "greater")$p.value
```

```
[1] 0.001416
```

```r
t.test(difference, mu = 0, conf.level = .99)$conf.int
```

```
[1] 0.3159 2.8441
attr(,"conf.level")
[1] 0.99
```

For a positive t statistic and a symmetric reference, the upper-tail p-value is half the two-sided value. The lower-tail value is its complement. Halving is inappropriate for an alternative opposite to the observed sign. Nonrejection is not proof of equality; significance does not specify a practical threshold or establish a causal explanation.

4 Basic Procedures for Means and a Proportion

4.1 Built-in records and explicit questions

The later examples use data already installed with R. They are separate illustrations, each with its own row unit and question. No supporting script or download is needed.

The official `mtcars` documentation describes 32 automobile models from 1973–74. `mpg` is miles per US gallon, `wt` is weight in thousands of pounds, and `am` codes automatic (0) or manual (1) transmission. These observational records are not a probability sample of all cars or a randomized transmission comparison.

```
cars <- mtcars
cars$transmission <- factor(cars$am, levels = c(0, 1),
                            labels = c("Automatic", "Manual"))
cars$mpg20 <- factor(cars$mpg >= 20,
                     levels = c(FALSE, TRUE),
                     labels = c("Below 20", "At least 20"))
mpg <- cars$mpg[!is.na(cars$mpg)]
head(cars[c("mpg", "wt", "transmission", "mpg20")])
```

```
                   mpg    wt transmission       mpg20
Mazda RX4         21.0 2.620       Manual At least 20
Mazda RX4 Wag     21.0 2.875       Manual At least 20
Datsun 710        22.8 2.320       Manual At least 20
Hornet 4 Drive    21.4 3.215    Automatic At least 20
Hornet Sportabout 18.7 3.440    Automatic    Below 20
Valiant           18.1 3.460    Automatic    Below 20
```

```
c(observed = length(mpg), missing = sum(is.na(cars$mpg)))
```

```
observed  missing
      32        0
```

The threshold 20 is a declared teaching definition, not a universal fuel standard. All 32 cars have the variables used here. Keeping an explicit missing-value check teaches how to retain the same denominator when another dataset is incomplete.

The mean and proportions describe these records directly. An interval or reference test adds a working process model. For example, a model of independent car outcomes is an additional assumption; running a function does not establish how the cars entered the file.

4.2 Known-sigma z and one-sample t

For independent normal outcomes with process mean μ and externally known SD σ, $SE(\bar{Y}) = \sigma/\sqrt{n}$ and $z = (\bar{y} - \mu_0)/SE$. Stipulate $\sigma = 5$ mpg and $\mu_0 = 20$ mpg only for this hypothetical calculation:

```
n_mpg <- length(mpg)
estimate_mpg <- mean(mpg)
z_se <- 5 / sqrt(n_mpg)
z <- (estimate_mpg - 20) / z_se
z_ci <- estimate_mpg + c(-1, 1) * qnorm(.975) * z_se
c(mean = estimate_mpg, SE = z_se, z = z)
```

```
   mean      SE       z
20.0906  0.8839  0.1025
```

```
z_ci
```

```
[1] 18.36 21.82
```

The recorded mean is 20.0906 mpg. The stipulated model gives SE .8839 and interval [18.3582, 21.8230] mpg. A known SD must come from an appropriate source; the observed file does not make 5 a known population SD.

When SD is estimated, replace σ by the sample SD s and use the t_{n-1} reference under the independent-normal model (Student, 1908).

```
mpg_se <- sd(mpg) / sqrt(n_mpg)
mpg_ci <- estimate_mpg + c(-1, 1) * qt(.975, n_mpg - 1) * mpg_se
one_t <- t.test(mpg, mu = 20)
c(mean = estimate_mpg, SD = sd(mpg), SE = mpg_se)
```

```
  mean     SD     SE 
20.091  6.027  1.065 
```

```r
mpg_ci
```

```
[1] 17.92 22.26
```

```r
one_t$statistic
```

```
      t 
0.08506 
```

```r
one_t$p.value
```

```
[1] 0.9328
```

Here $s = 6.0269$, $SE = 1.0654$, $t = .0851$, $df = 31$, and the 95% interval is [17.9177, 22.2636] mpg. The two-sided p value is .9328. This interval contains plausible process means under the model; it is not a range for individual cars. Its inclusion of 20 does not prove equality to 20.

Exact t calibration requires normal independent outcomes. Approximate use beyond that model depends on distribution shape, finite variance, influential values and the dependence structure, not a universal sample-size cutoff. The observed mean itself needs no such model to describe these cars.

4.3 Two independent means: Welch's comparison

For independent groups, define $\Delta = \mu_A - \mu_M$, here automatic minus manual mean mpg. The estimate and its SE are

$$\widehat{\Delta} = \bar{y}_A - \bar{y}_M, \qquad SE = \sqrt{s_A^2/n_A + s_M^2/n_M}.$$

Writing $a = s_A^2/n_A$ and $b = s_M^2/n_M$, the Welch degrees of freedom are

$$\nu = \frac{(a+b)^2}{a^2/(n_A - 1) + b^2/(n_M - 1)}.$$

```r
car_rows <- complete.cases(cars[c("mpg", "transmission")])
car_groups <- cars[car_rows, ]
auto <- car_groups$mpg[car_groups$transmission == "Automatic"]
manual <- car_groups$mpg[car_groups$transmission == "Manual"]
a <- var(auto) / length(auto)
b <- var(manual) / length(manual)
welch_se <- sqrt(a + b)
welch_df <- (a + b)^2 /
  (a^2 / (length(auto) - 1) + b^2 / (length(manual) - 1))
welch <- t.test(auto, manual)
c(n_auto = length(auto), n_manual = length(manual),
  auto_mean = mean(auto), manual_mean = mean(manual),
  difference = mean(auto) - mean(manual), SE = welch_se, df = welch_df)
```

```
     n_auto    n_manual   auto_mean manual_mean  difference          SE
     19.000      13.000      17.147      24.392      -7.245       1.923
         df
     18.332
```

```r
welch$conf.int
```

```
[1] -11.28  -3.21
attr(,"conf.level")
[1] 0.95
```

```r
welch$p.value
```

```
[1] 0.001374
```

There are 19 automatic and 13 manual cars, with means 17.1474 and 24.3923 mpg. Their difference is -7.2449 mpg, SE 1.9232, with approximate $df = 18.3323$ and interval $[-11.2802, -3.2097]$. The two-sided p value is .0014. The sign records the comparison's direction.

Welch's reference is approximate even for unequal-variance normal groups (Welch, 1947). It is the default two-sample `t.test()` calculation. `var.equal = TRUE` instead requests a pooled-variance procedure with $n_A + n_M - 2$ degrees of freedom. Neither option resolves dependence or confounding: these car models also differ in weight, engine and other features. The observed mpg gap does not isolate the effect of transmission.

The paired calculation in Section 3 instead uses within-person differences. Select paired or independent analysis from the design, not from which procedure gives a smaller p-value. Positive within-person covariance changes the variance of a difference, as Chapter 5 demonstrated.

4.4 One proportion: score test and Wilson interval

For an independent-Bernoulli working process with probability p, the estimate is $\hat{p} = s/n$. The score statistic for null p_0 is

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

Inverting $|z(p_0)| \leq z_{.975}$ gives the Wilson interval (Wilson, 1927):

$$\frac{\hat{p} + z^2/(2n)}{1 + z^2/n} \pm \frac{z}{1 + z^2/n}\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}.$$

```
successes <- sum(cars$mpg >= 20, na.rm = TRUE)
trials <- sum(!is.na(cars$mpg))
phat <- successes / trials
p0 <- .5
score_z <- (phat - p0) / sqrt(p0 * (1 - p0) / trials)
zcrit <- qnorm(.975)
wilson_center <- (phat + zcrit^2 / (2 * trials)) /
  (1 + zcrit^2 / trials)
wilson_half <- zcrit / (1 + zcrit^2 / trials) *
  sqrt(phat * (1 - phat) / trials + zcrit^2 / (4 * trials^2))
wilson_ci <- wilson_center + c(-1, 1) * wilson_half
score <- prop.test(successes, trials, p = p0, correct = FALSE)
c(successes = successes, trials = trials, proportion = phat, z = score_z)
```

```
 successes     trials proportion          z
   14.0000    32.0000     0.4375    -0.7071
```

```
wilson_ci
```

```
[1] 0.2817 0.6067
```

```
score$statistic
```

```
X-squared
      0.5
```

```r
score$p.value
```

```
[1] 0.4795
```

Fourteen of 32 recorded cars meet the definition: $\hat{p} = .4375$. The score statistic is $z = -.7071$; with continuity correction disabled, R reports $X^2 = z^2 = .5$, one df and $p = .4795$. The Wilson interval is [.2817, .6067]. The default continuity correction changes the statistic and interval, so declare the convention when reproducing a result.

This approximate reference requires an appropriate independent-trial model and adequate expected successes and failures. The Bernoulli model and the .50 reference are additional teaching assumptions. They do not turn the 32 recorded models into a random sample of all cars.

5 Association and Several Group Means

Chapter 5 supplied descriptive tables, correlations and group summaries. Here a reference model adds uncertainty to an explicitly stated question.

5.1 Correlation inference uses a different process target

Recreate the complete weight-mpg pairs introduced descriptively in Chapter 5:

```r
car_pairs <- cars[complete.cases(cars[c("wt", "mpg")]), ]
r <- cor(car_pairs$wt, car_pairs$mpg)
r
```

```
[1] -0.8677
```

The recorded $r = -.8677$ describes these 32 pairs. For an additional process correlation ρ, the zero-correlation statistic is

$$t = r\sqrt{\frac{n-2}{1-r^2}}, \qquad df = n - 2.$$

Its null reference is exact for independent bivariate-normal pairs. The approximate Fisher interval transforms $z_F = \operatorname{atanh}(r)$, uses $SE(z_F) = 1/\sqrt{n-3}$ and transforms back:

$$CI_\rho = \tanh\left(z_F \pm \frac{1.96}{\sqrt{n-3}}\right).$$

```r
n_r <- nrow(car_pairs)
fisher_ci <- tanh(atanh(r) + c(-1, 1) * qnorm(.975) / sqrt(n_r - 3))
cor_test <- cor.test(car_pairs$wt, car_pairs$mpg)
fisher_ci
```

```
[1] -0.9338 -0.7441
```

```r
cor_test$statistic
```

```
     t
-9.559
```

```r
cor_test$p.value
```

```
[1] 1.294e-10
```

The interval is $[-.9338, -.7441]$, $t = -9.5590$ with 30 df, and $p \approx 1.294 \times 10^{-10}$. These precise calculations do not verify bivariate normality, independent sampling or representation of all cars. When computing a whole covariance matrix, use a common row set: different pairwise missing-value subsets need not form a coherent joint covariance structure.

5.2 Categorical association: counts, residuals, and magnitude

Does the fraction with at least 20 mpg differ between automatic and manual cars in this file? Fix the observed rows before counting.

```r
categorical <- cars[complete.cases(cars[c("transmission", "mpg20")]), ]
counts <- table(categorical$transmission, categorical$mpg20)
counts
```

```
            Below 20 At least 20
  Automatic       15           4
  Manual           3          10
```

```r
rowSums(counts)
```

```
Automatic    Manual
       19        13
```

```r
prop.table(counts, margin = 1)
```

```
            Below 20 At least 20
  Automatic   0.7895      0.2105
  Manual      0.2308      0.7692
```

```r
chi_result <- chisq.test(counts, correct = FALSE)
chi_result$expected
```

```
            Below 20 At least 20
  Automatic   10.688       8.312
  Manual       7.312       5.688
```

```r
chi_result$residuals
```

```
            Below 20 At least 20
  Automatic    1.319      -1.496
  Manual      -1.595       1.808
```

```r
chi_result$stdres
```

```
            Below 20 At least 20
  Automatic    3.129      -3.129
  Manual      -3.129       3.129
```

```r
chi_result$statistic
```

```
X-squared
    9.791
```

```r
chi_result$p.value
```

```
[1] 0.001754
```

```r
cramers_v <- sqrt(as.numeric(chi_result$statistic) / sum(counts))
cramers_v
```

```
[1] 0.5531
```

Four of 19 automatic cars and ten of 13 manual cars meet the threshold: .2105 versus .7692. Under independence, the expected first-row/below-20 count is $19(18)/32 = 10.6875$, compared with 15 observed.

For cell counts O_{jk}, the fitted independence count, Pearson statistic, degrees of freedom and Cramér's magnitude measure are

$$E_{jk} = \frac{O_{j+}O_{+k}}{N}, \qquad X^2 = \sum_{jk} \frac{(O_{jk} - E_{jk})^2}{E_{jk}},$$
$$df = (J-1)(K-1), \quad V = \sqrt{\frac{X^2}{N \min(J-1, K-1)}}.$$

Here $X^2 = 9.7906$, one df, $p = .0018$ and $V = .5531$ (Cramér, 1946). Pearson residuals are $(O-E)/\sqrt{E}$. Standardized residuals additionally account for fitting the margins; they locate departures but are not a set of independent confirmatory tests. Row proportions retain the denominators needed to interpret the association.

The chi-square approximation needs independent units and adequate expected counts; the smallest expected cell here is 5.6875. The default two-by-two continuity correction is disabled explicitly. Fisher's alternative uses a conditional distribution given the margins under its null (Fisher, 1935):

```
fisher <- fisher.test(counts)
fisher$p.value
```

```
[1] 0.003289
```

The Fisher p value is .0033. Conditioning must fit the sampling or assignment question; Fisher's test is not a universal repair for dependence, selection or measurement problems. Neither result establishes that changing a car's transmission would cause it to cross the threshold.

5.3 Raw group means and one-way ANOVA

The official `ToothGrowth` documentation describes measurements from 60 guinea pigs. Each animal received one of three vitamin-C doses, .5, 1 or 2 mg/day, delivered as orange juice (`OJ`) or ascorbic acid (`VC`). `len` records odontoblast length; the help page does not specify a length unit, so we report the recorded length scale. This is not a human outcome or a measured before/after change. The short help page does not itself document a randomization protocol.

```r
growth <- ToothGrowth
growth$dose <- factor(growth$dose, levels = c(.5, 1, 2))
growth$supp <- factor(growth$supp, levels = c("OJ", "VC"))
table(growth$dose, growth$supp)
```

```
     OJ VC
 0.5 10 10
 1   10 10
 2   10 10
```

```r
dose_n <- table(growth$dose)
dose_mean <- tapply(growth$len, growth$dose, mean)
dose_sd <- tapply(growth$len, growth$dose, sd)
data.frame(dose = names(dose_n), n = as.integer(dose_n),
           mean = dose_mean, SD = dose_sd)
```

```
    dose  n  mean    SD
0.5  0.5 20 10.61 4.500
1      1 20 19.73 4.415
2      2 20 26.10 3.774
```

There are 20 animals per dose and ten per dose–supplement cell. The three raw dose means are 10.605, 19.735 and 26.100. Each averages equally across the two delivery methods. Treating `dose` as a factor fits separate means; it does not impose a linear dose–response relation.

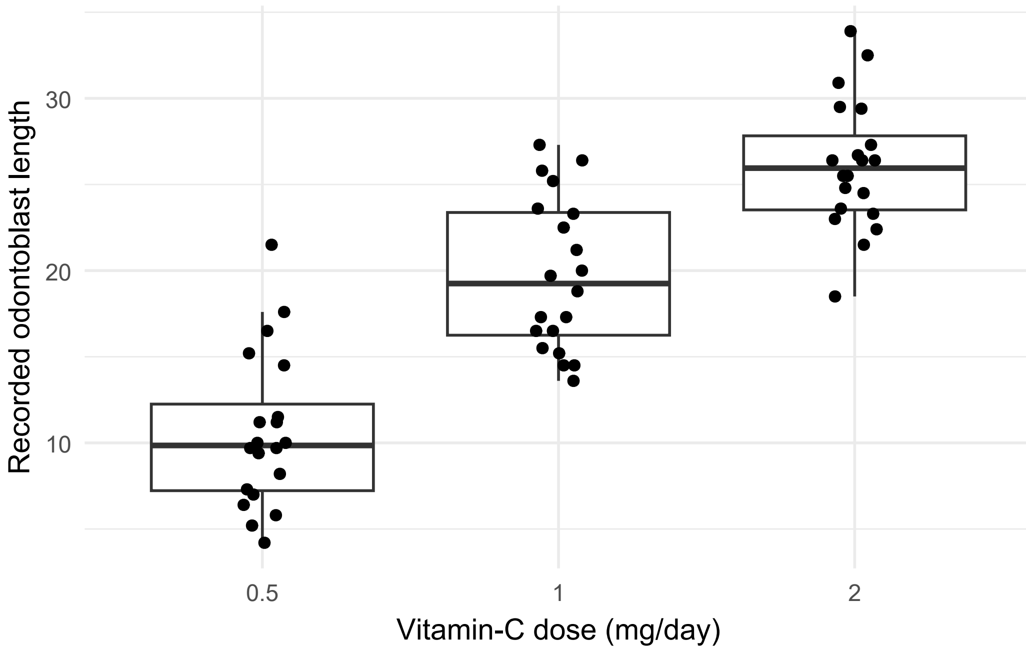

Figure 2: Individual lengths and distributions for the three dose groups.

For k groups and N records,

$$SS_B = \sum_j n_j(\bar{y}_j - \bar{y})^2, \quad SS_E = \sum_j \sum_i (y_{ij} - \bar{y}_j)^2, \quad F = \frac{SS_B/(k-1)}{SS_E/(N-k)}.$$

The omnibus null is equality of the three process means. Its exact normal reference requires independent errors, normal outcomes within groups and one common error variance. It does not require a pooled histogram to be normal.

```
dose_aov <- aov(len ~ dose, data = growth)
summary(dose_aov)
```

```
            Df Sum Sq Mean Sq F value  Pr(>F)
dose         2   2426    1213    67.4 9.5e-16 ***
Residuals   57   1026      18
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
dose_lm <- lm(len ~ dose, data = growth)
SSB <- sum(dose_n * (dose_mean - mean(growth$len))^2)
SSE <- sum(resid(dose_lm)^2)
MSE <- SSE / df.residual(dose_lm)
c(SSB = SSB, SSE = SSE, MSE = MSE)
```

```
 SSB  SSE  MSE 
2426 1026   18 
```

```r
anova(dose_lm)
```

```
Analysis of Variance Table

Response: len
          Df Sum Sq Mean Sq F value  Pr(>F)    
dose       2   2426    1213    67.4 9.5e-16 ***
Residuals 57   1026      18                    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
oneway.test(len ~ dose, data = growth)
```

```

	One-way analysis of means (not assuming equal variances)

data:  len and dose
F = 68, num df = 2, denom df = 38, p-value = 3e-13
```

Here $F = (2426.4343/2)/(1025.775/57) = 67.4157$, with $p < .001$ under that model. Welch's one-way extension gives $F = 68.4010$ with approximate denominator df 37.7433; it relaxes common variance through an approximate reference. Neither omnibus result selects the scientifically important contrast or validates the treatment protocol. Omitted delivery-method variation is one reason to examine the adjusted extension in Appendix C.

5.4 Magnitude, planned contrasts, and multiple comparisons

The one-way sample variation measures are

$$\eta^2 = \frac{SS_B}{SS_T}, \qquad \hat{\omega}^2 = \frac{SS_B - (k-1)MSE}{SS_T + MSE}, \qquad SS_T = SS_B + SS_E.$$

The omega estimate can be negative through sampling variation; report any zero-truncation convention explicitly. Neither measure is automatically a causal proportion of an outcome or a substantive importance threshold.

```
SST <- SSB + SSE
eta <- SSB / SST
omega <- (SSB - 2 * MSE) / (SST + MSE)
c(eta_squared = eta, omega_squared = omega)
```

```
  eta_squared omega_squared
       0.7029        0.6888
```

Here $\eta^2 = .7029$ and $\widehat{\omega}^2 = .6888$.

Declare a focused dose contrast before examining its results: the 2 mg/day mean minus the equally weighted average of the .5 and 1 mg/day means. Its weights are $\mathbf{c} = (-.5, -.5, 1)$.

$$\widehat{L} = \sum_j c_j \bar{y}_j, \quad SE(\widehat{L}) = \sqrt{MSE \sum_j c_j^2 / n_j}, \quad CI = \widehat{L} \pm t_{.975, N-k} SE(\widehat{L}).$$

```
weights <- c(-.5, -.5, 1)
contrast <- sum(weights * dose_mean)
contrast_se <- sqrt(MSE * sum(weights^2 / dose_n))
contrast_ci <- contrast + c(-1, 1) *
  qt(.975, df.residual(dose_lm)) * contrast_se
c(contrast = contrast, SE = contrast_se)
```

```
contrast       SE
  10.930    1.162
```

```
contrast_ci
```

```
[1]  8.604 13.256
```

```
TukeyHSD(dose_aov, "dose")
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = len ~ dose, data = growth)
```

```
$dose
        diff    lwr    upr p adj
1-0.5  9.130  5.902 12.358     0
2-0.5 15.495 12.267 18.723     0
2-1    6.365  3.137  9.593     0
```

The contrast is $26.100-(10.605+19.735)/2 = 10.930$, SE 1.1618, with interval [8.6036, 13.2564] on the recorded length scale. Equal group sizes make the chosen average equal the pooled mean of the two lower-dose groups here; those targets would differ if their sizes were unequal.

Tukey's output answers a different family: all three pairwise differences. Their estimates are 9.130, 15.495 and 6.365; the simultaneous intervals are [5.9018, 12.3582], [12.2668, 18.7232] and [3.1368, 9.5932]. Exact familywise coverage applies in the balanced normal common-variance case; `TukeyHSD()` uses an unequal-size adjustment otherwise. It does not account for searching across other outcomes or models. A numerical printed adjusted p value of zero is rounding, not an event with literally zero probability.

Keep the declared contrast and exploratory pairwise family distinct. Their positive findings concern these model-based comparisons; practical and causal interpretation also need the substantive measurement and design record.

6 Check Inference Through Repeated Sampling

6.1 Check decisions, not only whether the function ran

Inspect the IDs, measurement order and variables used to identify complete pairs. A wrong pairing changes the calculation. An unsupported independence assumption changes whether the SE has its proposed interpretation. A missing measurement restricts which people the description includes. Repeating a command cannot answer all three questions.

For `sleep`, the table establishes ten complete recorded pairs, and the arithmetic establishes their mean difference. The independent normal model supplies a conditional reference for uncertainty. The computation does not verify an unrecorded sampling or treatment protocol.

6.2 Simulate a known process

A simulation lets us choose the process and inspect its consequences. Generate independent normal differences with **known mean 1.5 hours** and **SD 1.2 hours**. Each generated study has $n = 10$; repeat it $B = 2000$ times. These are teaching constants, not estimated features or a reconstruction of the historical `sleep` study. The true mean is known in this experiment, so we can ask whether each generated interval contains it.

First follow one generated study: draw ten values, estimate its mean and SE, and form one interval. Check whether the declared mean 1.5 is inside.

```r
set.seed(60610)
one_sample <- rnorm(10, mean = 1.5, sd = 1.2)
one_mean <- mean(one_sample)
one_se <- sd(one_sample) / sqrt(10)
one_interval <- one_mean + c(-1, 1) * qt(.975, 9) * one_se
one_interval
```

```
[1] 1.021 2.212
```

```r
1.5 >= one_interval[1] & 1.5 <= one_interval[2]
```

```
[1] TRUE
```

The loop below repeats those same steps for fresh samples. Reset the seed before the full run so that its results do not depend on whether the one-study illustration was run first.

```r
set.seed(60610)
mu <- 1.5
sigma <- 1.2
n_sim <- 10
B <- 2000
sample_means <- numeric(B)
covered <- logical(B)
critical_sim <- qt(.975, df = n_sim - 1)
for (b in seq_len(B)) {
  sample_d <- rnorm(n_sim, mean = mu, sd = sigma)
  sample_means[b] <- mean(sample_d)
  sample_se <- sd(sample_d) / sqrt(n_sim)
  margin <- critical_sim * sample_se
  covered[b] <- abs(sample_means[b] - mu) <= margin
}
c(mean_of_means = mean(sample_means), empirical_SE = sd(sample_means),
  theoretical_SE = sigma / sqrt(n_sim), coverage = mean(covered))
```

```
 mean_of_means   empirical_SE theoretical_SE       coverage
        1.5015         0.3766         0.3795         0.9515
```

`numeric(B)` and `logical(B)` reserve one result per repetition. The loop draws a fresh sample, estimates its mean and SE, constructs its interval, then records whether it covers the known mean. A logical `TRUE` counts as one in `mean(covered)`. Repeating a calculation on one unchanged sample would not produce a sampling distribution.

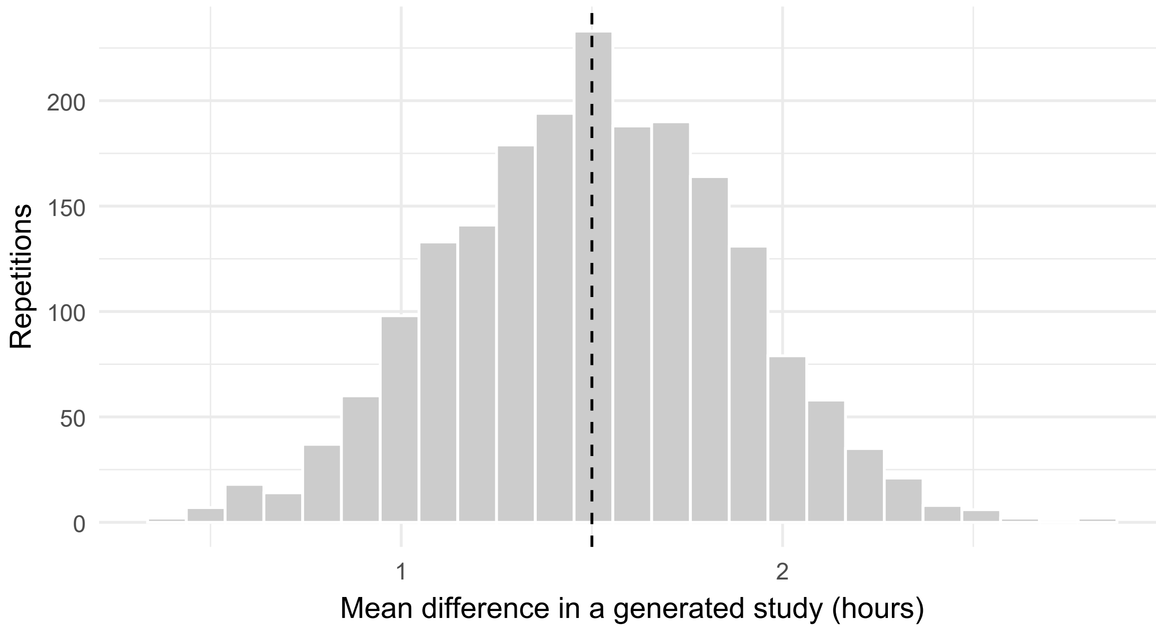

Figure 3: Sample means from the declared independent normal-differences process.

The distribution of 2,000 means approximates the **sampling distribution** under the chosen process. Its theoretical SE is $1.2/\sqrt{10} = .3795$ hours. Normality makes the usual t interval exact for this generated process. The observed coverage is 0.9515; finite simulations need not reproduce .9500 exactly.

Increasing n from 10 to 40 halves the theoretical SE to $1.2/\sqrt{40} = .1897$ hours. Increasing B leaves each study's information unchanged; it improves the numerical approximation from the simulation.

6.3 Separate three kinds of variability

Table 3
Study variation and simulation approximation are different.

Quantity	What varies?	Illustrative scale
Individual-difference SD	Observations within a generated study	1.2 hours
SE of the mean	Means across generated studies of size 10	$1.2/\sqrt{10} = .3795$ hours
Monte Carlo SE of coverage	Approximation from a finite number of simulations	About .0049 in probability units at $B = 2000$

If true coverage is $p = .95$, its Monte Carlo SE is

$$MCSE = \sqrt{\frac{p(1-p)}{B}} = \sqrt{\frac{.95(.05)}{2000}} = .0049.$$

For an unknown simulated probability, substitute the observed coverage. Increasing B from 2,000 to 8,000 halves this numerical SE, not the study mean's SE. The average simulated mean also has Monte Carlo uncertainty: about $.3795/\sqrt{2000} = .0085$ hours. A seed makes results repeatable; it does not make generating assumptions true for an actual study. Appendix A retains exact probabilities, skewed processes and further sample-size experiments.

7 Read the Answer and Prepare for Regression

7.1 A compact computational record

Read the calculation and its stated reasoning before evaluating a stronger claim. A concise report is:

> The ten complete person pairs had a mean recorded drug-2-minus-drug-1 difference of 1.58 hours of extra sleep, with SD 1.23 hours. Under a separately stated independent normal-differences model, the paired t procedure gave SE .389 hours and a 95% interval [.700, 2.460] hours for a process mean. The calculation does not itself establish the study's sampling, treatment-order or carryover conditions.

The inclusion code identifies the records, the dictionary explains the variables, and the returned object supplies the numerical quantities. The model declaration offers a reason for using the t

reference; locating that offered reason is separate from deciding whether it is adequate for the study. No full component map is needed for this computational lesson.

Separate quality check. "Drug 2 causes everyone to sleep 1.58 hours longer" changes an observed mean into a universal causal effect. It fails **Question alignment** and **Claim calibration**. The recorded mean remains checkable, but a population or causal answer requires a corresponding question and additional design support. Positive interval endpoints alone cannot supply that support.

7.2 A short bridge from a mean to a fitted line

For 2, 4 and 6, squared deviations total 8 around their mean of 4 and 11 around either 3 or 5. The mean minimizes the sum of squared deviations. With group A values 2 and 4 and group B values 6 and 10, the group means are 3 and 8. A fitted expression $3 + 5D$, where $D = 0$ for A and 1 for B, reproduces those means. The coefficient 5 is their observed difference.

These identities connect simple descriptive calculations to regression. They do not require Normal observations to hold; uncertainty for a model coefficient adds assumptions. Chapter 7 develops least squares, slope uncertainty and diagnostics, and Chapter 8 develops indicators, adjustment and coordinated prediction questions. Appendix B retains optional criterion and optimization details; Appendix C retains adjusted and factorial examples.

7.3 Summary and Reading Bridge

Name the estimand, relevant observations and dependence structure, then calculate an estimate, SE, interval or test under its stated reference. Read a returned number in that role. The same mean can be a fixed-record description or an estimate of a process parameter under an additional model.

The core methods are direct R calculations of familiar statistics. Their assumptions and interpretation are part of the analytical result. A sampling simulation illustrates a chosen process; it does not validate that process for historical records. Increasing sample size and increasing simulation repetitions improve different quantities.

The R t-test reference and local help such as `?prop.test`, `?cor.test` and `?aov` document the returned procedures. Chapter 7 extends the descriptive/inferential distinction to regression and Chapter 8 to multiple questions and model comparisons.

Exercises

Use the supplied summaries and reference values. Keep intermediate values unrounded; report SEs and intervals to three decimals and probabilities to four unless the question specifies otherwise.

Exercise 1: Preserve the pairs for inference (Foundational)

Four students' baseline scores are 40, 50, 60 and 70. Their corresponding follow-up scores are 46, 54, missing and 75. Calculate the complete-pair mean change and its sample SD. Under an independent normal-differences working model, calculate its estimated SE. Explain which observations the model concerns and why treating the measurements as independent groups would be a different calculation.

Exercise 2: Reproduce an interval (Foundational)

Twenty-five independent paired differences have mean 3.0 and SD 5.0. Under the normal-difference working model, use $t_{.975,24} = 2.064$. Calculate the estimated SE, the 95% interval and the standardized statistic for a zero process mean. Explain what the interval concerns and why it is not a range for individual differences.

Exercise 3: Match the procedure and output to the question (Foundational)

Study A measures the same 20 people twice. Study B compares two independent groups of 20 and 25 people without an equal-variance assumption. Name the appropriate mean-comparison procedures. A paired analysis returns estimate 2.4, SE 0.8, 95% interval [0.75,4.05] and $t = 3.0$. Explain each quantity. Name two additional pieces of information needed to assess a claim that the result is practically important and causal.

Exercise 4: More students or more simulations? (Foundational)

A simulation uses independent observations with SD 12, sample size 36 and 1,000 repetitions. Calculate the theoretical SE of a sample mean. If coverage is .95, calculate the Monte Carlo SE of the estimated coverage. What changes if sample size becomes 144? What changes if only the number of repetitions becomes 4,000? Report probabilities to four decimal places.

Exercise 5: Read a categorical comparison (Integrative)

Among 20 people in group A, 12 answer Yes and 8 No; among 20 in group B, 6 answer Yes and 14 No. Calculate the A-minus-B percentage-point difference and the four expected counts under independence. An uncorrected Pearson test reports $X^2 = 3.6364$, $df = 1$ and $p = .0565$. State the 5% decision and evaluate the statement "the groups have been proved identical."

Exercise 6: Separate an omnibus result from specific comparisons (Integrative)

A three-group ANOVA under an independent Normal common-variance model reports $F(2, 27) = 4.50$ and $p = .0206$. Two prespecified comparisons are B minus A and C minus A. Their simultaneous 95% intervals are [.2,3.8] and [-.1,6.1]. State what the omnibus result establishes at 5%. Does the evidence meet a rule requiring both comparisons to be positive? Explain why the second interval does not prove a zero difference.

Appendix A: Simulation Extensions

These optional examples deepen the distinction between study uncertainty and Monte Carlo approximation. Section 6 already supplies the chapter's essential simulation lesson.

Exact probabilities and Monte Carlo error

Simulation approximates $p = P(X \leq 4)$ under the binomial model. Let $I_b = 1$ when draw b meets that event and 0 otherwise. The Monte Carlo estimate is $\hat{p}_{MC} = B^{-1} \sum_b I_b$ with estimated numerical SE $\sqrt{\hat{p}_{MC}(1 - \hat{p}_{MC})/B}$.

```
set.seed(6062)
B_binomial <- 20000
binomial <- rbinom(B_binomial, size = 20, prob = .30)
exact <- pbinom(4, size = 20, prob = .30)
mc <- mean(binomial <= 4)
mcse <- sqrt(mc * (1 - mc) / B_binomial)
```

Table 4

A simulated probability and its numerical uncertainty.

Method	Probability	MC_SE
Exact model probability	0.2375	NA
Monte Carlo approximation	0.2389	0.00302

The exact probability is 0.23751; the Monte Carlo estimate is 0.23890, with MC SE 0.00302. At $B = 20{,}000$, the substitution is $\sqrt{\widehat{p}_{MC}(1-\widehat{p}_{MC})/20{,}000}$. Quadrupling B approximately halves this numerical SE. It does not improve the study's sample size, measurement, assignment, or the truth of the binomial assumptions.

Helpers, `replicate()`, and sample-size comparisons

A function can expose the sample size, repetition count, and generating rule as separate arguments. `replicate()` evaluates its expression repeatedly; the generator is called anew on every repetition, rather than taking the mean of one sample repeatedly.

```
simulate_means <- function(sample_size, repetitions,
    generator) {
  replicate(repetitions, mean(generator(sample_size)))
}
set.seed(6063)
sizes <- c(5, 20, 80)
sampling <- do.call(rbind, lapply(sizes,
    function(n_now) {
  data.frame(
    n = n_now,
    sample_mean = simulate_means(
      n_now, 5000, function(n) rlnorm(n, meanlog = 0, sdlog = .8)
    )
  )
}))
population_mean <- exp(.8^2 / 2)
population_sd <- sqrt((exp(.8^2) - 1) * exp(.8^2))
sampling_summary <- do.call(rbind, lapply(sizes,
    function(n_now) {
  estimates <- sampling$sample_mean[
    sampling$n == n_now]
  data.frame(n = n_now, Mean_estimate = mean(estimates),
             Empirical_SD = sd(estimates),
```

```
            Theoretical_SD = population_sd / sqrt(n_now),
            MC_SE_of_mean_estimate = sd(estimates) / sqrt(
    length(estimates)))
}))
```

Table 5

Observation variability, estimator variability, and Monte Carlo error.

n	Mean estimate	Empirical SD	Theoretical SD	MC SE
5	1.391	0.5947	0.5831	0.0084
20	1.369	0.2938	0.2916	0.0042
80	1.373	0.1467	0.1458	0.0021

The programmed lognormal population has mean $e^{.8^2/2}$ and variance $(e^{.8^2}-1)e^{.8^2}$. Its observation SD stays the same in all three conditions. The SD of its sample mean is $\sigma/\sqrt{n}$, so moving from $n = 5$ to 20 halves it. The last column instead measures numerical precision in the average of 5,000 simulated estimates. Increasing B improves that Monte Carlo approximation without making any individual simulated study more informative.

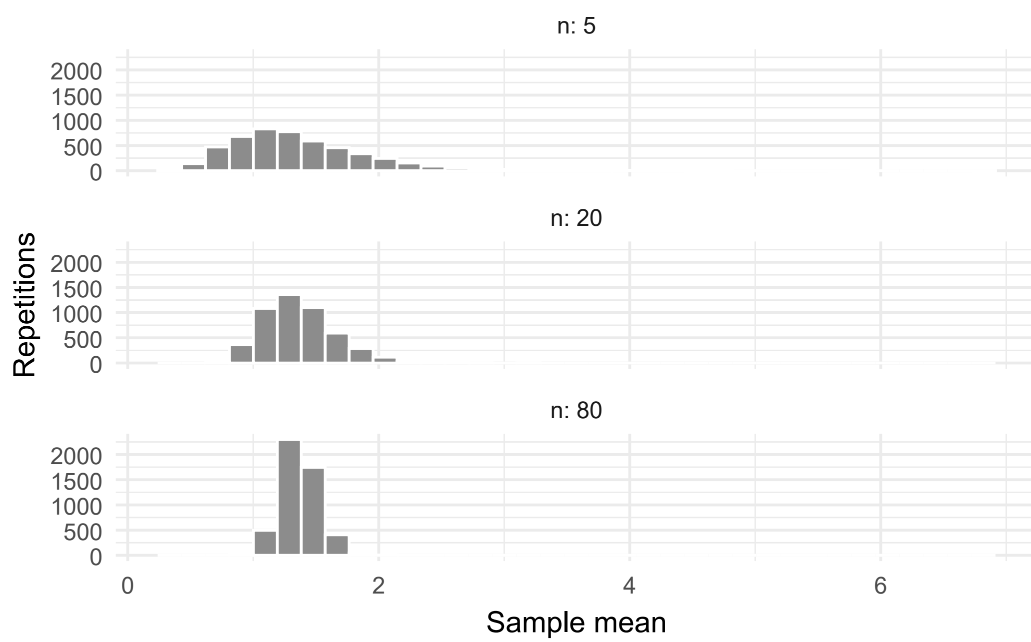

Figure 4: Sample means from the same right-skewed lognormal process at three sample sizes.

Appendix B: Estimation, Optimization, and Likelihood

A criterion defines the estimate

For values y_i, squared loss at candidate center m is $L(m) = \sum_i (y_i - m)^2$. Expanding around the mean gives

$$L(m) = \sum_i (y_i - \bar{y})^2 + n(m - \bar{y})^2.$$

The cross term vanishes because $\sum_i (y_i - \bar{y}) = 0$. Thus $\bar{y}$ is the unique minimizer when $n > 0$. Equivalently, $L'(m) = 2n(m - \bar{y})$ and $L''(m) = 2n > 0$. This identity does not require normality or sampling.

```
mpg <- mtcars$mpg
squared_loss <- function(m, y) sum((y - m)^2)
mean_fit <- optimize(squared_loss, interval = range(mpg), y = mpg)
c(numerical = mean_fit$minimum, arithmetic = mean(mpg),
  minimum_loss = mean_fit$objective)
```

```
   numerical   arithmetic minimum_loss
       20.09        20.09      1126.05
```

The numerical minimum is 20.0906 mpg, matching the arithmetic mean; loss is 1126.0472 squared mpg. Moving one mpg away raises it by $32(1)^2 = 32$. The search interval must contain the relevant minimum. An optimizer cannot find a value excluded by its constraints.

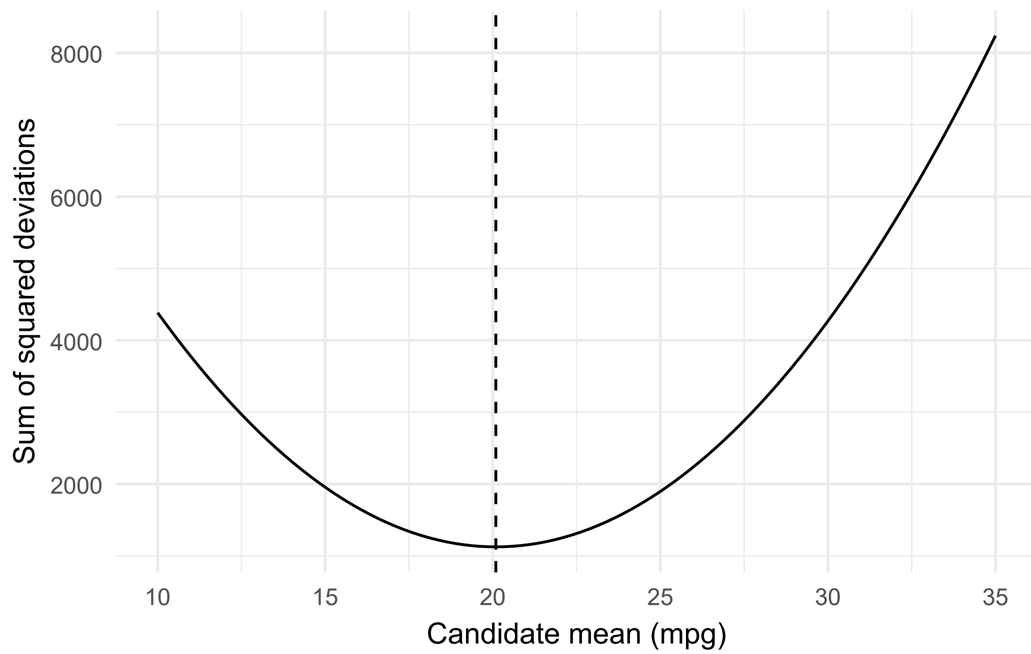

Figure 5: Squared loss is minimized at the recorded mean mpg.

Chapter 7 extends squared loss to a fitted line. The criterion defines what the estimate optimizes; it does not establish representation or causality.

A binomial likelihood and boundary solutions

For s successes in n independent Bernoulli trials with probability p,

$$L(p) = \binom{n}{s} p^s (1-p)^{n-s}, \qquad \ell(p) = \text{constant} + s\log p + (n-s)\log(1-p).$$

For $0 < s < n$, setting $s/p - (n-s)/(1-p) = 0$ gives $\hat{p} = s/n$. Use 14 of 32 cars reaching 20 mpg as an illustration of this separately stipulated likelihood. Counting those outcomes does not verify independent Bernoulli sampling.

```
successes <- sum(mtcars$mpg >= 20)
trials <- nrow(mtcars)
binomial_nll <- function(p, successes, trials) {
  -dbinom(successes, size = trials, prob = p, log = TRUE)
}
```

```r
prop_fit <- optimize(binomial_nll, interval = c(0, 1),
                     successes = successes, trials = trials)
c(numerical = prop_fit$minimum, proportion = successes / trials)
```

```
 numerical proportion
    0.4375     0.4375
```

Both values are .4375. The combinatorial term is included by `dbinom()` but does not affect which p maximizes the likelihood. If $s = 0$ or $s = n$, the maximum is at 0 or 1. Check endpoints explicitly in such cases; an interior-only search may omit the correct boundary solution. `log = TRUE` avoids taking the log of a probability that may already have underflowed.

Multiple parameters and convergence checks

Under a hypothetical independent normal model,

$$-\ell(\mu, \sigma) = \frac{n}{2} \log(2\pi) + n \log \sigma + \frac{\sum_i (y_i - \mu)^2}{2\sigma^2}.$$

Parameterizing $\sigma = e^\eta$ lets an unconstrained optimizer search in η while keeping scale positive. This illustrates likelihood fitting; it is not a claim that the recorded car outcomes follow that exact model.

```r
normal_nll <- function(par, y) {
  mu <- par[1]
  sigma <- exp(par[2])
  -sum(dnorm(y, mean = mu, sd = sigma, log = TRUE))
}
normal_fit <- optim(c(mean(mpg) - 1, log(sd(mpg))),
                    normal_nll, y = mpg, method = "BFGS", hessian = TRUE)
second_fit <- optim(c(mean(mpg) + 1, log(sd(mpg) / 2)),
                    normal_nll, y = mpg, method = "BFGS")
c(numerical_mean = normal_fit$par[1],
  numerical_scale = exp(normal_fit$par[2]),
  arithmetic_mean = mean(mpg),
  closed_form_scale = sqrt(mean((mpg - mean(mpg))^2)))
```

```
   numerical_mean   numerical_scale   arithmetic_mean closed_form_scale
           20.091             5.932            20.091             5.932
```

```r
normal_fit$convergence
```

```
[1] 0
```

```r
normal_fit$value - second_fit$value
```

```
[1] -5.988e-07
```

```r
eigen(normal_fit$hessian)$values
```

```
[1] 64.0006  0.9094
```

The mean is approximately 20.0906 and normal MLE scale 5.9320. The latter divides squared deviations by n, whereas `sd()` divides by $n-1$. The fits return convergence code zero and nearly identical objective values. A zero code reports an algorithm's stopping criterion. Compare starting values, the returned objective and an analytic answer where available. Positive Hessian eigenvalues support local curvature, not a universal proof of global identification or scientific model adequacy.

Extended practice: an estimand and its criterion (Integrative)

Plot squared loss for a quantitative variable and compare its minimum with the arithmetic mean. Deliberately narrow the search interval to exclude the mean, then explain why the numerical result changes. For binomial counts, compare an interior case with zero successes and all successes, checking the boundary likelihoods. Distinguish the scientific question, chosen criterion and numerical search.

Appendix C: Adjusted and Factorial Model Extensions

These optional extensions develop the group comparisons in Section 5. They are useful when adjustment or effect variation belongs to the question; they are not prerequisites for the chapter's main paired-analysis route.

Indicator regression reproduces the same fitted group means

With .5 mg/day as reference, the intercept is its mean; the other two coefficients are the 1-minus-.5 and 2-minus-.5 differences. Adding each coefficient to the intercept reconstructs its group mean.

```
coef(dose_lm)
```

```
(Intercept)       dose1       dose2
      10.61        9.13       15.49
```

```
dose_levels <- levels(growth$dose)
new_doses <- data.frame(dose = factor(dose_levels, levels = dose_levels))
predict(dose_lm, new_doses)
```

```
    1     2     3
10.61 19.73 26.10
```

The coefficients are 10.605, 9.130 and 15.495, and the fitted group means are 10.605, 19.735 and 26.100. The intercept-plus-indicators model and one-way ANOVA fit the same space, so fitted values, residuals and the classical joint F agree. This is a computational identity; it adds no new design evidence. Chapter 8 develops the regression algebra further.

Adjustment and balanced comparisons

In `ToothGrowth`, supplement is a second recorded treatment dimension, not a school block or a before/after occasion. An additive model assumes a common dose comparison within OJ and VC. A marginal dose mean instead averages over the supplements observed at that dose. Here the balanced cell counts give equal supplement weights at every dose. Consequently the raw and additive-model contrasts have the same point estimate; their SEs use different residual variances. This equality need not hold with unequal supplement compositions.

```
growth <- ToothGrowth
growth$dose <- factor(growth$dose, levels = c(.5, 1, 2))
growth$supp <- factor(growth$supp, levels = c("OJ", "VC"))
supp_fit <- lm(len ~ supp, data = growth)
add_fit <- lm(len ~ supp + dose, data = growth)
coef(add_fit)
```

```
(Intercept)      suppVC       dose1       dose2
      12.45       -3.70        9.13       15.50
```

```r
anova(supp_fit, add_fit)
```

```
Analysis of Variance Table

Model 1: len ~ supp
Model 2: len ~ supp + dose
  Res.Df  RSS Df Sum of Sq    F Pr(>F)
1     58 3247
2     56  820  2      2426 82.8 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The additive intercept is 12.455. The VC-minus-OJ coefficient is -3.700; dose coefficients are 9.130 and 15.495 relative to .5 mg/day. The dose block comparison gives $F(2, 56) = 82.8109$, $p < .001$. This is a joint test of dose terms, not the test of the single high-dose contrast.

For the 2 mg/day mean minus the equally weighted average of .5 and 1, the coefficient weights are $(0, 0, -.5, 1)$ in the displayed order.

```r
L <- c(0, 0, -.5, 1)
adjusted <- sum(L * coef(add_fit))
adjusted_se <- sqrt(drop(t(L) %*% vcov(add_fit) %*% L))
adjusted_ci <- adjusted + c(-1, 1) *
  qt(.975, df.residual(add_fit)) * adjusted_se
c(contrast = adjusted, SE = adjusted_se)
```

```
contrast       SE
  10.930    1.048
```

```r
adjusted_ci
```

```
[1]  8.83 13.03
```

The adjusted contrast is 10.9300, SE 1.0482, interval [8.8301, 13.0299]. Compare the raw one-way SE 1.1618 and interval [8.6036, 13.2564]. The smaller residual variance after including supplement changes the precision under the additive model; the data have not gained more animals. With exact additivity, a common within-supplement dose contrast also equals a contrast standardized to any common supplement weights. Whether a common contrast is useful depends on how the comparison varies by supplement.

Six cells and explicit model comparisons

Two supplements and three doses form six observed cells, with ten animals per cell. Inspect the counts and means before fitting an interaction:

```
cell_n <- with(growth, table(supp, dose))
cell_mean <- with(growth, tapply(len, list(supp, dose), mean))
cell_n
```

```
    dose
supp 0.5  1  2
  OJ  10 10 10
  VC  10 10 10
```

```
cell_mean
```

```
     0.5     1     2
OJ 13.23 22.70 26.06
VC  7.98 16.77 26.14
```

The OJ means are 13.23, 22.70 and 26.06; the VC means are 7.98, 16.77 and 26.14. The high-minus-low dose difference is 12.83 under OJ and 18.16 under VC. Interaction concerns such variation in comparisons, not simply different cell means.

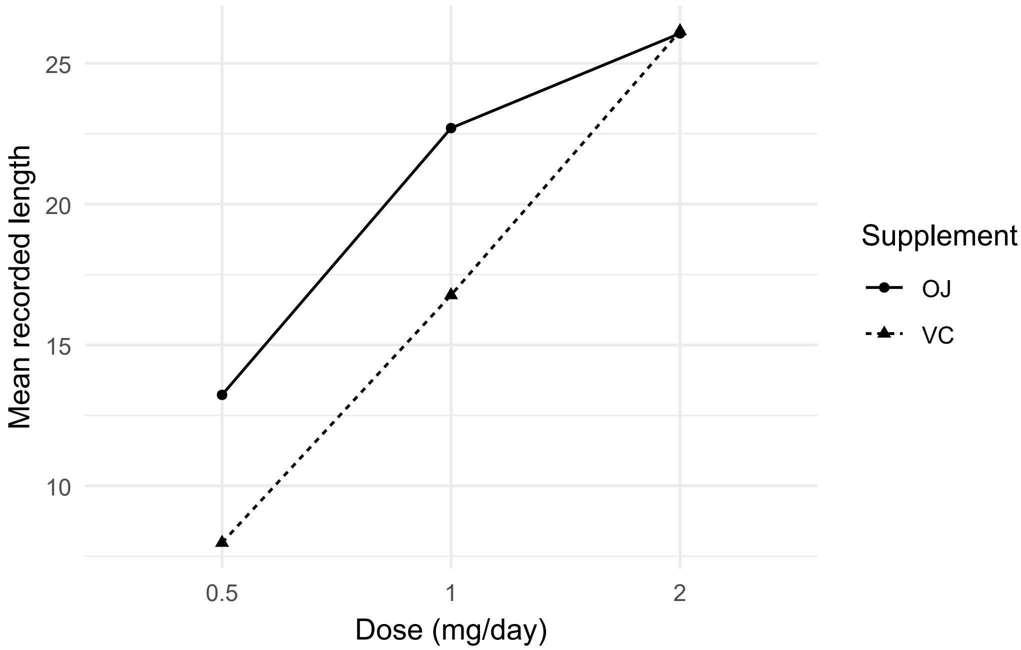

Figure 6: Mean recorded lengths for the six dose–supplement combinations.

The additive model has four coefficients: intercept, one supplement and two dose terms. The full product model has six cell means, adding $(2-1)(3-1) = 2$ interaction parameters and leaving $60 - 6 = 54$ residual df.

For nested models, let $SS_{\text{added}} = RSS_R - RSS_F$ and let q be the number of extra parameters. The ordinary nested-model statistic is

$$F = \frac{SS_{\text{added}}/q}{RSS_F/df_F}.$$

```r
full_fit <- lm(len ~ supp * dose, data = growth)
anova(add_fit, full_fit)
```

```
Analysis of Variance Table

Model 1: len ~ supp + dose
Model 2: len ~ supp * dose
  Res.Df RSS Df Sum of Sq    F Pr(>F)
1     56 820
2     54 712  2       108 4.11  0.022 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
rss_add <- sum(resid(add_fit)^2)
rss_full <- sum(resid(full_fit)^2)
df_full <- df.residual(full_fit)
interaction_F <- ((rss_add - rss_full) / 2) / (rss_full / df_full)
interaction_p <- pf(interaction_F, 2, df_full, lower.tail = FALSE)
c(F = interaction_F, p = interaction_p)
```

```
      F       p
4.10699 0.02186
```

The extra SS is 108.319, full-model residual SS 712.106, and $F = (108.319/2)/(712.106/54) = 4.1070$, $p = .0219$. Under the working model, this is evidence against the additive mean restriction. Interpret the dose profiles before deciding whether a single average comparison is adequate. The exact reference uses independent normal, common-variance errors and the specified nested mean spaces (Kutner et al., 2005).

One can also report several components against a deliberately common full-model residual denominator. Name that choice explicitly:

```r
dose_only <- lm(len ~ dose, data = growth)
component_ss <- c(
  sum(resid(dose_only)^2) - rss_add,
  sum(resid(supp_fit)^2) - rss_add,
  rss_add - rss_full
)
component_df <- c(1, 2, 2)
component_F <- (component_ss / component_df) / (rss_full / df_full)
data.frame(component = c("Supplement after dose", "Dose after supplement",
```

```
                    "Interaction"),
        df = component_df, SS = component_ss, F = component_F,
        p = pf(component_F, component_df, df_full, lower.tail = FALSE))
```

```
              component df     SS      F         p
1 Supplement after dose  1  205.4 15.572 2.312e-04
2 Dose after supplement  2 2426.4 92.000 4.046e-18
3           Interaction  2  108.3  4.107 2.186e-02
```

The component F values are 15.5720, 92.0000 and 4.1070. The first two use a denominator from the full product model; they therefore differ from ordinary comparisons ending at the additive model. Only the last is the same nested interaction test just shown. In this balanced design, supplement and dose terms are orthogonal. With unequal cells, sequential sums of squares can depend on term order, so name the compared spaces instead of treating any row labelled a main effect as interchangeable.

None of these model identities supplies missing assignment documentation or extends animal results to people. Statistical fit, precision, substantive importance and causal interpretation remain separate questions.

Appendix D: Extended Practice and Synthesis

These optional tasks combine procedure selection, simulation, and bounded multi-procedure interpretation.

Extended practice: simulation questions (Foundational)

Write distinct mass/density, cumulative-probability, quantile, and simulation questions for a binomial and a normal distribution. Approximate one exactly computable probability at two values of B and report both MC SEs. Then compare means at two values of n while holding the generator fixed. Explain separately what varies among observations, among study estimates, and because the simulation used finitely many repetitions. Do not describe this lognormal example as the source of the built-in observed records.

Extended practice: match procedure to target (Integrative)

For a known-SD mean, unknown-SD mean, independent mean difference, paired difference, and one proportion, name the estimand and repeated world before the function. Reproduce estimate, SE, interval, and null standardization; state whether the reference is exact or approximate and which conditions make that statement true. Do not describe a stipulated SD as empirically known.

For two quantitative variables, fix a complete-pair sample, plot it, and reproduce covariance, Pearson r, and the Fisher interval. For a two-way categorical table, show observed and expected counts, denominators, standardized residuals, chi-square, and Cramer's V. Explain when Fisher's conditional test addresses the same intended null and what neither test can say about design or causation.

A bounded multi-procedure reading and extended practice (Integrative)

An extended report might state: "In the 60 recorded guinea pigs, dose means were 10.605, 19.735 and 26.100 on the recorded length scale. The declared high-dose-minus-average-lower-dose contrast was 10.930, with a common-variance working-model interval [8.604, 13.256]. An additive supplement-adjusted model gave the same estimate and interval [8.830, 13.030]. The dose profiles differed by supplement; a nested interaction comparison gave $F(2, 54) = 4.107$."

Read each quantity with its question, denominator and working model. Equal raw and adjusted estimates here follow from balance, not a universal rule. The interaction result invites interpretation of conditional dose comparisons; it does not replace that interpretation with a significance label. The dataset help page does not supply a threshold of substantive importance or a complete randomization record.

For practice, reproduce the raw means, one-way ANOVA, magnitude measures, declared contrast and Tukey family. Explain their different questions. Verify that indicator regression returns the same fitted means. Then reproduce the six cell means, additive contrast and nested interaction statistic. Compare the residual denominators used for the different reported F values.

Complete a short computational record using one actual built-in dataset or explicitly labelled supplied summary. State the question, row unit, variables, calculation, result, assumptions and bounded interpretation. Include one useful display or compact table. Read what the calculation establishes before judging one consequential limitation. Finish by explaining what a later regression model would add to the chosen question.

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